

Introduction

Sports are beneficial for children of all ages. Children can enjoy physical activities by participating in sports and games. It's crucial that children's motor development aligns with their requirements.

Common Postural Deformities

Postural deformities are conditions where there is an abnormal alignment of the body segments. This section covers common deformities related to the spine and lower limbs, along with their causes and corrective measures.

Spinal Curvature

These deformities are related to the spine and can be caused by carrying excessive weight or weak muscles.

- **Kyphosis:** An abnormal outward curve of the spine (hunchback), most common in older women.
 - **Causes:** Malnutrition, heavy loads, weak muscles.
 - **Remedies:** Use a pillow under the back while sleeping, perform Dhanurasana (bow pose).
- **Lordosis:** An abnormal inward curve of the spine (swayback), especially in the lumbar region.
 - **Causes:** Improper muscle development, obesity.
 - **Remedies:** Perform sit-ups, perform Halasana (plow pose).
- **Scoliosis:** A lateral (sideways) curvature of the spine.
 - **Causes:** Underdeveloped legs, paralysis.
 - **Remedies:** Bending exercises, swimming using breaststroke techniques.

Lower Limb Deformities

- **Knock Knees (Genu Valgum):** A condition where the knees knock or touch each other when standing, while the ankles remain apart.
 - **Causes:** Lack of balanced diet, especially deficiency in Vitamin D, calcium, and phosphorus.
 - **Remedies:** Perform Padmasana (lotus pose), Gomukhasana (cow face pose).
- **Bow Legs (Genu Varum):** The opposite of knock knees, where there is a wide gap between the knees when standing with feet together.
 - **Causes:** Deficiency of calcium, phosphorus, and bone issues.
 - **Remedies:** Vitamin D supplements, balanced diet.
- **Round Shoulder:** A postural deformity where the shoulders become rounded and appear to be bent forward.
 - **Causes:** Heredity, tight clothing.
 - **Remedies:** Perform Chakrasana (wheel pose), Dhanurasana (bow pose).

Exercises for Postural Deformities

- **Kyphosis:** Examples include chest stretches, back extensions, and exercises strengthening the upper back.
- **Lordosis:** Exercises focusing on core strength, pelvic tilts, and hamstring stretches.
- **Scoliosis:** Specific bending, stretching, and strengthening exercises to address the lateral curve. Pilates for Scoliosis is also a method.
- **Knock Knees:** Exercises to strengthen inner thigh muscles and stretch outer thigh muscles.
- **Bow Legs:** Exercises to strengthen outer thigh muscles and stretch inner thigh muscles, like Garudasana (eagle pose) and Ardha Matsyendrasana (half lord of the fishes pose).
- **Round Shoulders:** Exercises like Chakrasana, Dhanurasana, and Gomukhasana to improve shoulder posture.

Special Consideration for Sportswomen

Sportswomen need to consider specific physiological and psychological factors, including menarche and menstrual dysfunction.

Menarche

- **Definition:** A girl's first menstrual period, typically occurring between ages 9 and 16. Influenced by biological, genetic, and environmental factors.
- **Sports Context:** Sportswomen need proper nutrition during this period to ensure normal bodily function.

Menstrual Dysfunction

- **Definition:** Abnormal bleeding and irregular menstrual cycles (cycle lasting 22-35 days, period 3-7 days). Symptoms include headache, back pain, fatigue, cramps.
- **Impact:** Affects physical performance and psychological well-being. Emphasizes the need for cleanliness and hygienic habits.

Female Athlete Triad

The Female Athlete Triad is a syndrome of three interrelated conditions often seen in teenage or adult female athletes who don't meet their energy requirements, leading to undernourishment and severe performance impact.

1. **Low Energy Availability (Disordered Eating):** Insufficient caloric intake relative to energy expenditure.
2. **Menstrual Disturbance (Amenorrhea):** Absence or irregularity of menstrual periods.
3. **Bone Loss (Osteoporosis):** Weakening of bones due to decreased bone density.

Osteoporosis

- **Definition:** Weakening of bones due to loss of bone density and improper bone formation, leading to fractures and injuries. Characterized by low estrogen and minerals, making bones porous and brittle.
- **Causes:** Low estrogen levels, poor nutrition (especially low calcium), imbalance of thyroxine hormone (hyperthyroidism).
- **Preventive Measures:**
 - Consume calcium and protein-rich foods.
 - Ensure adequate Vitamin D intake (sunlight or supplements).
 - Avoid smoking and alcohol.
 - Regular check-ups.
 - Perform proper physical activity.

Amenorrhea

- **Definition:** Absence or irregularity of menstrual periods caused by intense exercise and insufficient calorie intake, leading to hormonal imbalances.
- **Symptoms:** Hair loss, headache, vision changes, facial hair development.
- **Causes:** Low estrogen levels, poor nutrition (especially low calcium), imbalance of thyroxine hormone (hyperthyroidism).
- **Preventive Measures:**
 - Consume calcium and protein-rich foods.
 - Ensure adequate Vitamin D intake (sunlight or supplements).
 - Avoid smoking and alcohol.
 - Regular check-ups.
 - Perform proper physical activity.

Eating Disorder

- **Definition:** Taking very few calories, leading to deficiency of essential nutrients and lack of energy. Related to mental illness, affecting physical and mental health. Athletes trying to lose weight may not eat enough fat and iron, leading to disorders.
 - **Anorexia Nervosa:** Obsession with weight, distorted body image, fear of being overweight. Maintaining below-normal weight through starvation or excessive exercise.
 - **Bulimia Nervosa:** Binge eating followed by intentional vomiting (purging), excessive exercising, or fasting to avoid weight gain. A serious, potentially life-threatening disorder.